

TABLEAU

Module 7 :- Data Visualization — Part II

Advanced & Custom Charts

Charts Covered in This Module

Topic 1	Scatter Plot — show relationship between two measures
Topic 2	Bubble Chart — scatter plot with a third measure as bubble size
Topic 3	Maps — show data on a geographic map of India
Topic 4	Dumbbell Chart — compare two values for the same item
Topic 5	Sankey Diagram — show flow of data between two groups
Topic 6	Word Cloud — show importance using text size

Same Cricket Tournament Dataset • Simple English • Step-by-Step Every Chart

Dataset Reminder — Cricket Tournament 2024

We use the same 20-row cricket dataset from Module 6. Here is a quick reminder of what the data contains so you can follow along with every chart in this module.

Column	What It Means	Example Values
Team	Name of the cricket team	Mumbai Lions, Chennai Kings, Delhi Eagles...
City	Home city of the team	Mumbai, Chennai, Delhi, Bangalore...
Match	Which match was played	Match 1, Match 3, QF 1, SF 1, Final
Runs	How many runs the team scored	165, 187, 210, 245...
Wickets	How many wickets the team lost	1, 3, 6, 9...
Fours	Number of four-run hits	12, 18, 24, 28...
Sixes	Number of six-run hits	5, 9, 13, 16...
Win/Loss	Did the team win or lose?	Win or Loss
Target	Runs they needed to win	165, 175, 198, 232...
Phase	Tournament stage	League or Playoff

Extended Data for Maps Topic

For the Maps topic (Topic 3), we will use City and Team together. Each team's home city is a real Indian city — Mumbai, Delhi, Chennai, Bangalore, Kolkata, Pune, Hyderabad, Jaipur (Rajasthan). Tableau can plot these on an India map automatically.

Topic 1 — Scatter Plot

● Scatter Plot

Shows the *RELATIONSHIP* between two numbers — do they move together or not?

1.1 What is a Scatter Plot? — Simple Explanation

A Scatter Plot puts two numbers on a chart — one on the X axis (horizontal) and one on the Y axis (vertical). Each team (or item) becomes ONE DOT on the chart. Where the dot lands depends on both numbers together.

Simple example: *If a team scored 210 runs and lost only 2 wickets, their dot is placed at X=210, Y=2. A team that scored 165 runs and lost 9 wickets sits at X=165, Y=9. You now see all teams as dots — and the pattern of those dots tells you something important.*

Cricket analogy — Why use a Scatter Plot?

You want to know: do teams that score more runs also lose fewer wickets? In other words, is high scoring linked to better batting control? A scatter plot shows this instantly. If most dots go from bottom-left to top-right — yes, they are related. If dots are scattered randomly everywhere — no clear relationship.

1.2 Understanding Correlation — 3 Patterns

Correlation means — do two things move together? There are 3 patterns you will see:

Pattern	What It Means	Cricket Example
Positive Correlation	When one goes UP, the other also goes UP. Dots form a line going from bottom-left to top-right.	<i>More fours hit → more runs scored. Makes sense!</i>
Negative Correlation	When one goes UP, the other goes DOWN. Dots form a line going from top-left to bottom-right.	<i>More wickets lost → fewer runs (team collapsed)</i>
No Correlation	Dots are scattered randomly. No clear pattern. The two numbers are NOT related.	<i>Number of sixes vs match number — no pattern</i>

1.3 The Data — Runs vs Wickets per Innings

Here is what each innings looks like when plotted on a scatter plot:

Team	Runs (X axis)	Wickets Lost (Y axis)	Result	Quadrant
Mumbai Lions (Final)	245	1	Win	High Runs, Low Wickets — Best performance
Mumbai Lions (SF)	230	2	Win	High Runs, Low Wickets — Dominant
Chennai Kings (QF)	215	4	Win	High Runs, Medium Wickets
Mumbai Lions (QF)	221	3	Win	High Runs, Low Wickets
Chennai Kings (M2)	203	4	Win	Good Runs, Low Wickets
Mumbai Lions (M5)	210	2	Win	High Runs, Very Low Wickets
Hyderabad Hawks	195	3	Win	Good Runs, Low Wickets
Kolkata Tigers	165	9	Loss	Low Runs, High Wickets — Struggled

1.4 Step-by-Step — Build a Scatter Plot

1	Open a new blank worksheet in Tableau
2	In the Marks card dropdown, keep it as 'Automatic' — Tableau will use circles by default for scatter plots
3	Drag 'Runs' to the Columns shelf — this becomes the X axis (horizontal)
4	Drag 'Wickets' to the Rows shelf — this becomes the Y axis (vertical)
5	Right now you see only ONE dot — because Tableau has aggregated everything into one total. We need to separate by team.
6	Drag 'Team' to the Detail shelf on the Marks card — now each team becomes its own dot
7	You should now see multiple dots — one per team (or per innings if duplicates exist)
8	Drag 'Team' also to the Color shelf — each team's dot gets a different colour
9	Drag 'Team' also to the Label shelf — team names appear next to each dot
10	Change the aggregation: right-click 'SUM(Runs)' on Columns → Dimension (to plot individual rows, not totals)
11	Do the same for Wickets on Rows → right-click → Dimension
12	Now every single innings is a separate dot — you see the full scatter pattern

1.5 Adding a Trend Line

A Trend Line is a straight line drawn through the middle of all the dots. It shows the general direction of the relationship — going up (positive correlation) or going down (negative correlation).

1	After building your scatter plot, go to the top menu: Analytics tab (next to the Data tab on the left panel)
2	In the Analytics panel, you will see 'Trend Line' under the 'Model' section
3	Drag 'Trend Line' onto the chart — drop it on 'Linear' (straight line)
4	A line appears through the dots with a shaded band around it (confidence interval)
5	Hover over the trend line — a tooltip shows the formula: $\text{Wickets} = a + b \times \text{Runs}$ and the R-squared value
6	R-squared close to 1.0 = strong relationship. Close to 0 = no real relationship
7	To remove it: right-click the trend line → Remove Trend Lines

1.6 Adding Reference Lines — Average Lines

Reference lines divide your scatter plot into 4 quadrants — this is called Quadrant Analysis. You draw one average line on X and one on Y. Each quadrant means something different.

1	Go to the Analytics panel (left side, Analytics tab)
2	Drag 'Reference Line' onto the chart → drop it on 'Table' for the X axis first
3	In the dialog: set Value = SUM(Runs) → Average. Label = 'Avg Runs'. Click OK
4	A vertical reference line appears at the average runs value
5	Drag another 'Reference Line' → drop it on the Y axis this time
6	Set Value = SUM(Wickets) → Average. Label = 'Avg Wickets'. Click OK
7	Now your chart has 4 quadrants — like a 2×2 grid over your dots

What each quadrant means:

Quadrant	Runs	Wickets	Meaning	Teams Here
Top-Right	Above avg	Above avg	Many runs BUT lost many wickets — scraped through	Delhi, Rajasthan
Top-Left	Below avg	Above avg	Low runs AND lost many wickets — very poor performance	Kolkata
Bottom-Right	Above avg	Below avg	HIGH runs, FEW wickets — BEST performance	Mumbai, Chennai
Bottom-Left	Below avg	Below avg	Low runs but controlled wickets — steady but not enough	Pune, Bangalore

Topic 2 — Bubble Chart



Bubble Chart

A Scatter Plot where each dot has a *SIZE* — showing *THREE* measures at once

2.1 What is a Bubble Chart? — Simple Explanation

A Bubble Chart is exactly like a Scatter Plot — dots on X and Y axes. The only difference is: the *SIZE* of each dot represents a *THIRD* number. So instead of two measures, you show three measures in one single chart.

Simple example: *X axis = Runs scored. Y axis = Wickets lost. Bubble SIZE = Number of Sixes hit. A team with a big bubble hit lots of sixes. A team with a small bubble relied on singles and twos. You see three stories at once.*

Cricket analogy

Mumbai Lions scored 245 runs (X), lost only 1 wicket (Y), and hit 16 sixes (big bubble). Kolkata Tigers scored 165 runs (X), lost 9 wickets (Y), and hit only 5 sixes (small bubble). Just by looking at the bubble — big, bottom-right dot vs small, top-left dot — you see the quality difference between the two innings.

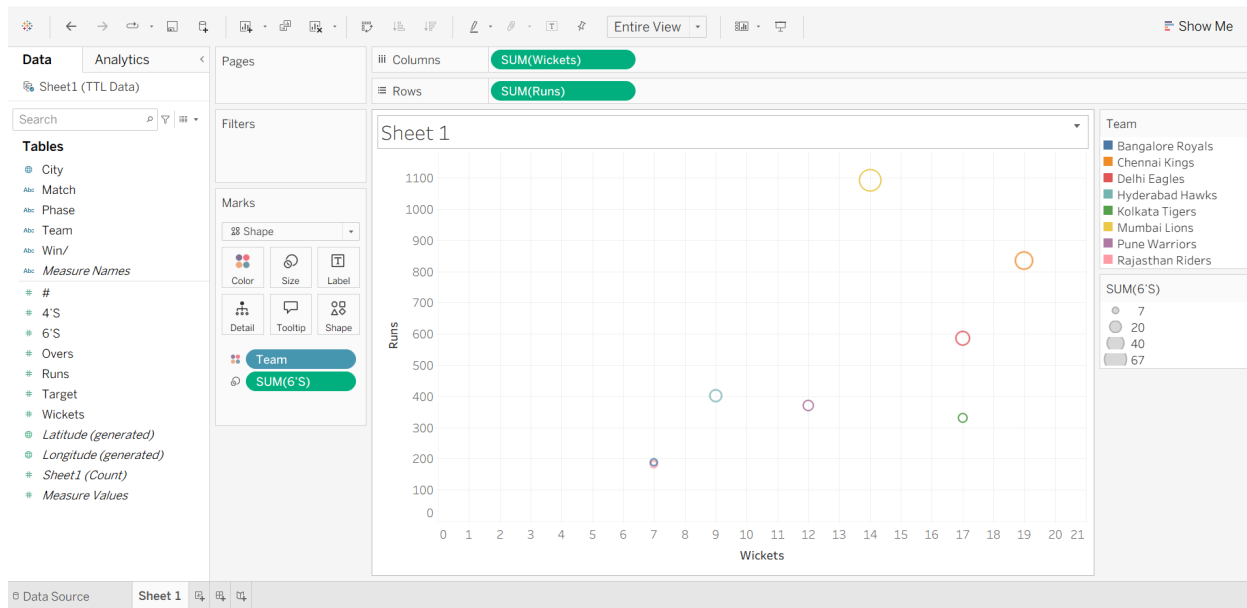
2.2 Scatter Plot vs Bubble Chart

Scatter Plot	Bubble Chart
Shows 2 measures (X and Y)	Shows 3 measures (X, Y, and Size)
All dots are the same size	Dot size shows a third value
Good for finding correlation	Good for finding correlation + magnitude
Cleaner and simpler to read	Slightly more complex — big bubbles can overlap
Use when you have 2 key numbers	Use when size/volume is also important

2.3 Step-by-Step — Build a Bubble Chart

We will plot: X = Runs, Y = Wickets, Bubble Size = Sixes, Colour = Team

1	Start with the Scatter Plot you built in Topic 1 — Runs on Columns, Wickets on Rows, Team on Color and Label
2	Now go to the Marks card and find the 'Size' shelf
3	Drag 'Sixes' from the Data panel to the 'Size' shelf on the Marks card
4	The dots immediately change size — teams that hit more sixes now have bigger dots
5	Adjust the size range: click the 'Size' shelf → move the slider to make the size difference more visible (spread them out)
6	Drag 'Sixes' also to the Tooltip shelf so when you hover over a bubble, it shows the sixes count
7	If bubbles overlap too much: go to the Marks card dropdown → change from 'Automatic' to 'Circle' → reduce opacity to 70% using the Color shelf → Opacity slider
8	To add a size legend: check that the Sixes size legend appears on the right side of the view. If not, go to Analysis menu → Legends → Size Legend
9	Title: 'Batting Performance — Runs vs Wickets (Bubble = Sixes)'



2.4 Reading the Bubble Chart — What to Look For

What You See	What It Means
Big bubble in bottom-right corner	High runs, few wickets, many sixes — DOMINANT innings
Small bubble in top-left corner	Low runs, many wickets, few sixes — COLLAPSE
Big bubble in top-right	Many sixes hit but also many wickets lost — risky batting
Small bubble in bottom-right	High runs without many sixes — solid, singles-based batting
Two bubbles close together	Those two teams had very similar performances

When NOT to use Bubble Charts

Avoid bubble charts when: (1) you have more than 15-20 items — too many bubbles overlap and become unreadable. (2) The size differences are very small — bubbles all look the same size. In these cases, use a regular scatter plot with colour, or a heat map instead.

Topic 3 — Maps (Geographic Visualization)



Maps

Show data ON A MAP — location tells the story when geography matters

3.1 What is a Map Chart? — Simple Explanation

A Map chart places your data ON AN ACTUAL MAP. Instead of bars or dots on a blank canvas, your data appears on top of a real geographic map — India map, world map, or city map. The position of each mark is determined by the location field in your data.

Simple example: *Each cricket team has a home city. Mumbai Lions → Mumbai. Chennai Kings → Chennai. If you show Total Runs by City on a map, each city appears as a dot or filled region on the India map — and its size or colour shows the runs. You instantly see which region of India has the strongest cricket teams.*

Cricket analogy

Your dataset has a 'City' column with Mumbai, Delhi, Chennai, Bangalore, Kolkata, Pune, Hyderabad, Jaipur. Tableau knows these are Indian cities. It plots them on the India map automatically. A large dot over Mumbai means Mumbai Lions scored a lot of runs. A small dot over Jaipur (Rajasthan Riders) means fewer runs.

3.2 Symbol Map vs Filled Map

Symbol Map (Dot/Bubble Map)	Filled Map (Choropleth Map)
Shows a DOT or circle at each location	Fills the entire region with colour
Dot size = one measure, Dot colour = another	Colour intensity = the measure value
Works for cities, specific locations	Works for states, countries, regions
Good when exact location matters	Good when regional patterns matter
Cricket: dot per city, size = team's runs	Cricket: fill each state by total runs from that state
Easier to build — Tableau does it automatically	Requires matching state/country names exactly

3.3 Setting Up Geographic Roles

Before building a map, you need to tell Tableau that your 'City' column contains geographic data. This is called setting a Geographic Role.

1	In the Data panel, find the 'City' field
2	Right-click 'City' → Geographic Role → City (Tableau marks the field with a small globe icon — it now knows this is a location)
3	Do the same for any other geographic fields — right-click 'Team' or a 'State' field → Geographic Role → State/Province if needed
4	If Tableau cannot find a city (shows a small warning icon), it means the city name is not recognised — we will fix this in step 3.6

3.4 Build a Symbol Map — Runs by City

A dot on each city's location, sized by total runs scored by the team from that city.

1	Make sure 'City' has a Geographic Role set to City (from step 3.3 above)
2	Double-click 'City' in the Data panel — Tableau AUTOMATICALLY creates a map! Latitude goes to Rows, Longitude goes to Columns
3	You should see dots appearing over Indian cities on the map
4	Drag 'SUM(Runs)' to the Size shelf on the Marks card — dots get bigger for cities with more runs
5	Drag 'Team' to the Color shelf — each team's city dot gets a different colour
6	Drag 'Team' to the Label shelf — team names appear next to each city dot
7	Drag 'SUM(Runs)' to the Tooltip shelf — when you hover over a dot, the runs total appears
8	To zoom into India: use the map zoom controls (+ and – buttons on the map, or scroll with mouse)
9	Title: 'Total Runs by Team Home City — India Map'

What you will see on the map:

City	Team	Total Runs	Expected Dot Size
Mumbai	Mumbai Lions	1,093	Largest dot — tournament winners
Chennai	Chennai Kings	616	Large dot
Delhi	Delhi Eagles	585	Large dot
Hyderabad	Hyderabad Hawks	402	Medium dot
Kolkata	Kolkata Tigers	331	Small dot
Pune	Pune Warriors	370	Small dot
Bangalore	Bangalore Royals	189	Tiny dot
Jaipur	Rajasthan Riders	182	Smallest dot

3.5 Build a Filled Map — Wins by State

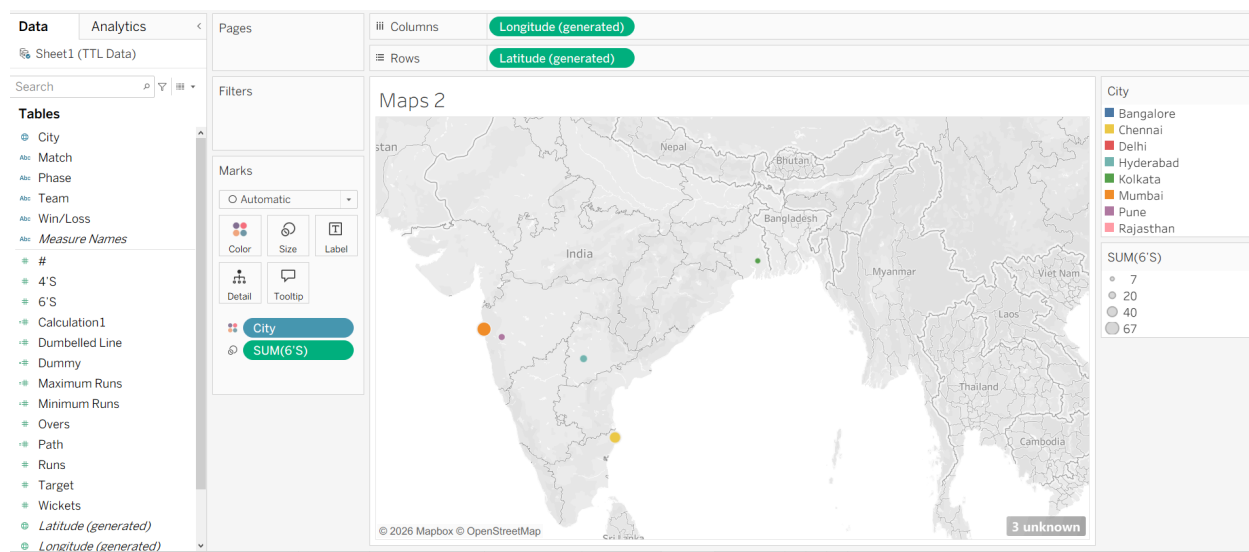
The entire state region is filled with a colour. Darker = more wins from teams in that state.

1	In the Data panel, right-click 'City' → Geographic Role → State/Province (We treat city as state here for simplicity, or add a separate State column)
2	Double-click 'City' — the map appears with dots as before
3	In the Marks card dropdown, change the mark type from 'Automatic' to 'Filled Map'
4	The map now shows filled state regions instead of dots
5	Drag 'SUM(Runs)' to the Color shelf — states fill with colour based on total runs
6	Click the Color shelf → Edit Colors → choose a sequential palette (e.g. Blue: light to dark)
7	Darker blue = more runs from teams in that state. Lighter blue = fewer runs
8	Title: 'Total Runs by State — Filled Map'

3.6 Fixing Unrecognised Locations

Sometimes Tableau cannot find a city on the map and shows a small warning icon at the bottom of the screen saying 'X unknown locations'. Here is how to fix it:

1	Look at the bottom of the map — click the warning message 'X unknown'
2	A dialog opens showing which location names Tableau could not match
3	For each unrecognised name, type the correct spelling or match it to a known location
4	Example: 'Bengaluru' might not be found — type 'Bangalore' which Tableau recognises
5	Click OK — the dots now appear on the correct locations
6	Alternative fix: right-click the city name in the Data panel → Aliases → type the standard name Tableau accepts



Topic 4 — Dumbbell Chart



Dumbbell Chart

Compares TWO values for the same item — the gap between the dots is the story

4.1 What is a Dumbbell Chart? — Simple Explanation

A Dumbbell Chart shows TWO dots connected by a line for each item (category). One dot = first value. Other dot = second value. The line connecting them shows the gap or distance between the two values. It looks like a dumbbell weight — hence the name.

Simple example: *You want to compare the minimum and maximum temperature in different cities for one month. Each city gets two dots — one for the coldest day (left dot) and one for the hottest day (right dot). A line connects them. The longer the line, the bigger the temperature range in that city.*

Cricket analogy

Each team's LOWEST score and HIGHEST score across all their matches. Mumbai Lions scored as low as 187 (Match 1) and as high as 245 (Final). The dumbbell shows both dots and a line connecting them. A long line = big range in performance (inconsistent). A short line = consistent performance throughout the tournament.

4.2 The Data — Min and Max Runs by Team

Team	Lowest Score	Highest Score	Gap (Range)	What It Means
Mumbai Lions	187	245	58	Consistent improvement — grew stronger each match
Chennai Kings	198	215	17	Very consistent — narrow range
Delhi Eagles	175	232	57	Big range — inconsistent, had both bad and good games
Hyderabad Hawks	182	207	25	Fairly consistent
Kolkata Tigers	165	166	1	Very consistent but consistently low
Pune Warriors	171	199	28	Decent range
Bangalore Royals	189	189	0	Played only one match
Rajasthan Riders	182	182	0	Played only one match

4.3 Step-by-Step — Build a Dumbbell Chart

We build this using a Dual Axis chart with Circle marks and a Line connecting them.

First create two calculated fields:

Calculated Field 1: Min Runs
Formula: `MIN([Runs])`

Calculated Field 2: Max Runs
Formula: `MAX([Runs])`

1	Drag 'Team' to the Rows shelf
2	Drag 'Min Runs' to the Columns shelf — a bar appears for each team's minimum score
3	Drag 'Max Runs' ALSO to the Columns shelf — a second chart appears next to the first
4	Right-click 'MAX(Max Runs)' on the Columns shelf → Dual Axis (Both measures now share the same axis)
5	Right-click the right axis → Synchronize Axis (so both axes match)
6	In the Marks card, you now have TWO mark cards — one for Min Runs, one for Max Runs
7	Click the 'Min Runs' mark card → change type to 'Circle'. Set colour to Blue. Increase size using Size slider.
8	Click the 'Max Runs' mark card → change type to 'Circle'. Set colour to Orange. Same size as Min.
9	Now we need the connecting LINE: create a third calculated field called 'Dumbbell Line' Formula: <code>MIN([Runs])</code> (same as Min Runs — this creates the base of the line)
10	Drag 'Dumbbell Line' to the Columns shelf → Dual Axis again
11	Change this mark card type to 'Bar'. Set the bar colour to a light grey.
12	The grey bar stretches from 0 to Min Runs. We want it from Min to Max — so: right-click the x-axis → Edit Axis → set Fixed Start to a number just below the minimum score (e.g. 150)
13	Now the grey bar stretches between the two circles — creating the dumbbell look
14	Sort by Max Runs descending: right-click Team on Rows → Sort → Field → Max Runs → Descending
15	Title: 'Lowest vs Highest Score by Team — Performance Range'

Topic 5 — Sankey Diagram

~ Sankey Diagram

Shows FLOW between two groups — how much goes from each source to each destination

5.1 What is a Sankey Diagram? — Simple Explanation

A Sankey Diagram shows how things FLOW from one group to another. On the left side are SOURCE categories. On the right side are DESTINATION categories. Thick curved bands connect them — the THICKER the band, the MORE that flows between those two things.

Cricket analogy

Left side = Teams. Right side = Tournament Phases (League, Playoff). The band thickness shows how many RUNS flowed from each team into each phase. Mumbai Lions have a thick band to Playoff (they played many playoff matches and scored a lot). Rajasthan Riders have only a thin band to League (only 1 match, then out). You see which teams dominated which phases.

5.2 Understanding the Data Flow

Team (Source)	Phase (Destination)	Runs (Flow Thickness)
Mumbai Lions	League	397 runs (Match 1 + Match 5)
Mumbai Lions	Playoff	696 runs (QF + SF + Final)
Chennai Kings	League	401 runs (Match 2 + Match 5)
Chennai Kings	Playoff	433 runs (QF + SF)
Delhi Eagles	League	353 runs (Match 1 + Match 6)
Delhi Eagles	Playoff	232 runs (Final only)
Kolkata Tigers	League	331 runs (Match 3 + Match 6)
Kolkata Tigers	Playoff	0 — did not qualify

5.3 Step-by-Step — Build a Sankey Diagram

A Sankey Diagram in Tableau requires a technique called data densification — we create extra rows of data to draw the curved bands. This is the most complex chart in this module. Follow each step carefully.

Part A — Prepare the data structure:

1	Create a new sheet in your Excel file called 'Sankey_Data' with these columns: Team Phase Runs Fill it with the 8 rows from the table above (each team + phase combination)
2	Add a column called 'T' (for path parameter) with values 0 and 1 for each row. Duplicate every row — one copy with T=0 (left side) and one with T=1 (right side) You now have 16 rows (8 original × 2)
3	Connect Tableau to this Sankey_Data sheet

Part B — Create the required calculated fields:

Field 1: Sigmoid Curve (draws the S-shaped bands)

Formula: $1 / (1 + \text{EXP}(-(6.5 * [T] - 3.25)))$

Field 2: Running Total Left (stacks items on left side)

Formula: `RUNNING_SUM(SUM([Runs]))`

Field 3: Running Total Right (stacks items on right side)

Formula: `RUNNING_SUM(SUM([Runs]))`

Field 4: Rank (for positioning)

Formula: `RANK_UNIQUE(SUM([Runs]))`

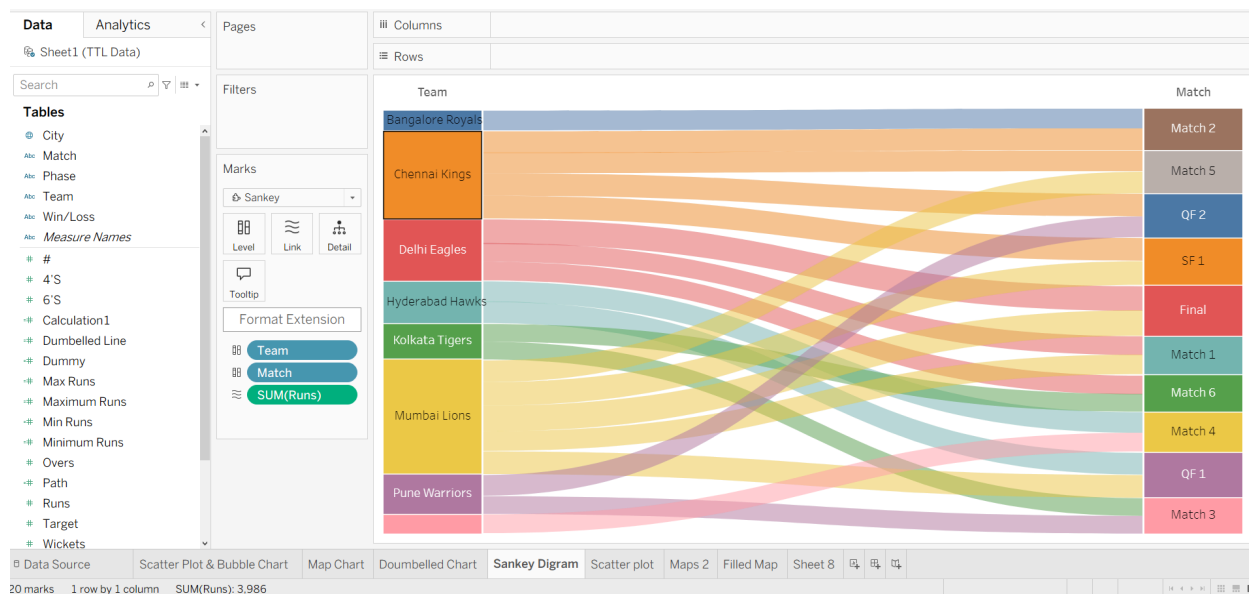
Part C — Build the chart:

1	Drag 'Sigmoid Curve' to the Columns shelf
2	Drag 'Running Total Left' to the Rows shelf
3	In the Marks card, change the type to 'Line'
4	Drag 'Team' to the Detail shelf — one line per team
5	Drag 'Phase' to the Color shelf — League and Playoff get different colours
6	Drag 'T' to the Path shelf in the Marks card — this tells Tableau the order of the curve points

7	Right-click 'Running Total Left' on Rows → Edit Table Calculation → set Compute Using = Team
8	Drag 'Runs' to the Size shelf — band thickness = runs scored
9	The curved S-shaped bands now flow from Team (left) to Phase (right)
10	Format: remove gridlines (Format menu → Lines → set all to None). Remove axis headers. Add a title.

Sankey Is Complex — That is OK!

Sankey diagrams are one of the hardest charts to build in Tableau. If you cannot build it perfectly right away, that is completely normal. Many experienced Tableau users use a Sankey template or download pre-built Sankey workbooks from Tableau Public to start from. Focus on understanding WHAT it shows — the how-to-build will come with practice



Topic 6 — Word Cloud

🧩 Word Cloud

Shows words in different SIZES — bigger word = more important or more frequent

6.1 What is a Word Cloud? — Simple Explanation

A Word Cloud is a visual where words are displayed in different sizes. A word that appears more often, or has a higher value, is shown BIGGER. A word that appears less often is shown SMALLER. It gives a quick visual summary of which items are most prominent.

Simple example: *You have a list of all words in a book. The word 'cricket' appears 500 times, 'ball' appears 300 times, 'wicket' appears 100 times. In a word cloud, 'cricket' is the biggest word in the centre, 'ball' is slightly smaller, 'wicket' is even smaller. You instantly see what the book is mostly about.*

Cricket analogy

We have 8 teams in the tournament. Each team scored a different total of runs. Mumbai Lions scored the most (1,093). Rajasthan Riders scored the least (182). In a Word Cloud, 'Mumbai Lions' appears as the BIGGEST text. 'Rajasthan Riders' appears small. The word sizes tell you tournament dominance without any bars or axes at all.

6.2 When to Use a Word Cloud

✅ Word Cloud Works Well	❌ Word Cloud Does NOT Work Well
When you want a quick visual summary	When exact values matter — hard to read precise sizes
For presentations and dashboards as eye-catch	For serious analysis — too imprecise
When you have 10–50 items to display	When you have 2–4 items — use a bar chart
When differences in size are very large	When all items are similar size — looks flat
Team names sized by runs or sixes	Comparing values where differences are small (e.g. 180 vs 190)

6.3 Step-by-Step — Build a Word Cloud (Teams by Runs)

Team names will appear in different sizes based on total runs scored. Biggest = most runs.

1	Open a new blank worksheet
2	In the Marks card dropdown, change the chart type to 'Text' (This is the mark type that creates Word Clouds in Tableau)
3	Drag 'Team' to the Label shelf on the Marks card — all team names appear on screen
4	Drag 'SUM(Runs)' to the Size shelf on the Marks card — teams with more runs appear in larger text
5	Drag 'Team' also to the Color shelf — each team name gets a different colour
6	Drag 'SUM(Runs)' also to the Tooltip shelf — hovering over a team name shows their run total
7	Adjust size range: click the Size shelf → drag the slider to make the size difference more dramatic (The default size range is often too small — spread it out so the difference is obvious)
8	To make it look like a proper word cloud: go to Format → Alignment → set Horizontal = Centre and Vertical = Middle
9	Expand the chart window to fill the view — the words will spread out and overlap less
10	Title: 'Tournament Word Cloud — Team Names Sized by Total Runs'

6.4 Word Cloud Variation — Sixes by Team

Instead of Runs, size the team names by Sixes hit. This gives a different word cloud — teams that relied on power hitting appear biggest.

1	Follow all steps from 7.3 but in Step 4, drag 'SUM(Sixes)' to the Size shelf instead of Runs
2	The word cloud now shows a different pattern — teams with big-hitting style appear larger
3	Notice: Mumbai Lions still dominate (67 sixes). But the relative sizes of other teams change compared to the Runs-based word cloud
4	This is a great way to show two different stories from the same data — just by changing the Size measure

Module 7 — Quick Reference Cheat Sheet

Advanced Chart — When to Use Each

Chart	Use When	Cricket Example	Key Shelves
Scatter Plot	You want to see the RELATIONSHIP between 2 numbers	Runs (X) vs Wickets (Y) — do they correlate?	X measure → Columns, Y measure → Rows, Category → Color/Detail
Bubble Chart	Same as scatter but SIZE of dot also matters	Runs vs Wickets, bubble = Sixes count	Same as Scatter + 3rd measure → Size shelf
Symbol Map	Location data + one value to show as dot size	Team home cities, dot size = total runs	Double-click geo field, Size = measure
Filled Map	Location data shown as filled regions with colour	State filled by total runs from that state	Marks = Filled Map, measure → Color
Dumbbell Chart	Compare MIN and MAX (or before/after) per item	Each team's lowest vs highest score	Min + Max → Columns → Dual Axis, mark = Circle
Sankey Diagram	Show FLOW between two groups by volume	Teams (left) → Phase (right), band = runs	Sigmoid curve + Running Total + T on Path
Word Cloud	Show many items with size showing importance	Team names sized by runs or sixes	Marks = Text, Category → Label, measure → Size

Master Chart Selector — Module 6 + Module 7 Combined

Your Question	Best Chart	Module
Compare values across categories	Horizontal Bar Chart	6
Show composition breakdown of a total	Stacked / 100% Stacked Bar	6
Show trend over time or sequence	Line Chart	6
Compare two measures on same chart	Dual Axis Chart	6
Show simple part-of-whole (≤5 items)	Pie or Donut Chart	6

Your Question	Best Chart	Module
Compare many items by size (6+ items)	Treemap	6
Show actual value vs target/benchmark	Bullet Chart	6
Compare values in a matrix (2 dimensions)	Heat Map or Highlight Table	6
Find relationship between 2 numbers	Scatter Plot	7
Find relationship + show a 3rd measure	Bubble Chart	7
Show data by geographic location	Symbol Map or Filled Map	7
Compare two values per item (min vs max)	Dumbbell Chart	7
Show flow between two groups	Sankey Diagram	7
Show items by importance/frequency	Word Cloud	7

End of Module 7 — Data Visualization Part II

Practice: Try building the Scatter Plot first, then the Bubble Chart, then the Word Cloud.